



Internship Experience UK | BRIGHT
NETWORK

Technology - Sector Skills Project

You only need to complete one sector skills project.

Please choose either the Technology or Engineering sector skills project.

You will only be able to submit one project.

The logo for Bright Network is centered within a large white circle. The word 'BRIGHT' is written in a large, orange, outlined sans-serif font. Below it, the word 'NETWORK' is written in a smaller, dark grey, solid sans-serif font.

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Scenario & Client Issue

Scenario

- **You are a Product Analyst at EcoCart, a rapidly growing retail tech startup**
- **Their core product is a digital grocery list and barcode scanning app that help users track their purchases**

Client Issue

- **The cost-of-living crisis and rising environmental concerns have shifted consumer priorities.**
- **They are increasingly frustrated about what happens after their shopping - users report buying groceries, forgetting what is in their fridge and throwing away expired food**



Vision & Goal



- The EcoCart founders want to build a brand-new feature powered by **Generative AI**.
- The goal is to allow users to:
 - Scan their paper or digital grocery receipts;
 - have the app automatically estimate expiration dates;
 - and suggest personalised, budget-friendly recipes based on what needs to be eaten first.

Your Role

The engineering team is ready to build, but we need a **clear product strategy**.

The founders are unsure how to structure the user experience, what high-level tech is required, or how to handle sensitive purchasing data.

Your task is to:

**Define the
Minimum
Viable Product
(MVP)**

**Present a
strategic
plan to the
board**



User Research & Insights

To help you design the right solution, our UX team has provided a recent snapshot of **user feedback** and **two primary target personas**.

User Feedback:



I love tracking my spending, but my biggest issue is wasting food. I buy spinach, push it back to the back of the fridge and find it a week later ruined. Help!



Can you add a feature that tells me what to cook? Groceries are so expensive right now, I can't afford to buy specific ingredients. I just want to use what I already have.



I'd love to scan my receipts, but I'm worried about my data. Does EcoCart sell purchasing habits to advertisers?

User Research & Insights

Target Personas



Persona A: Sarah
(32, Working Parent)

Time-poor and budget-conscious. She wants to stretch her weekly grocery budget as far as possible and needs quick, 30-minute recipes that her kids will actually eat.



Persona B: Liam
(20, University Student)

Highly eco-conscious and hates the environmental impact of food waste. However, he is a beginner cook and lacks the culinary intuition to invent a meal out of random leftover ingredients.

Deliverables required

You must submit a **Product Strategy Portfolio** consisting of three distinct artifacts.

**Product
Requirements
Document
(PRD) Excerpt**

**UX
Wireframes
(Image / PDF)**

**Strategic
Pitch Deck
(Maximum 10
Slides)**



Artifact 1 & 2

Artifact 1: PRD Excerpt

Outline the exact requirements for the engineering and design teams.

- **User Stories:** Write at least 3 agile user stories using the standard format ("As a [user type], I want to [action] so that [benefit]"). Include 1-2 bullet points of Acceptance Criteria for each.
- **Success Metrics:** Identify 3 specific Key Performance Indicators (KPIs) EcoCart should track to measure if this feature is successfully reducing food waste and engaging users.

Artifact 2: UX Wireframes (Image / PDF)

Create a low-to-mid-fidelity visual mockup of the new feature. You can use free tools (like Figma, Balsamiq, or Canva) or neat, hand-drawn sketches.

You must include at least **3 screens** showing the user journey:

- The receipt scanning interface.
- The digitized "inventory" view showing expiration warnings.
- The AI-generated recipe suggestion screen.

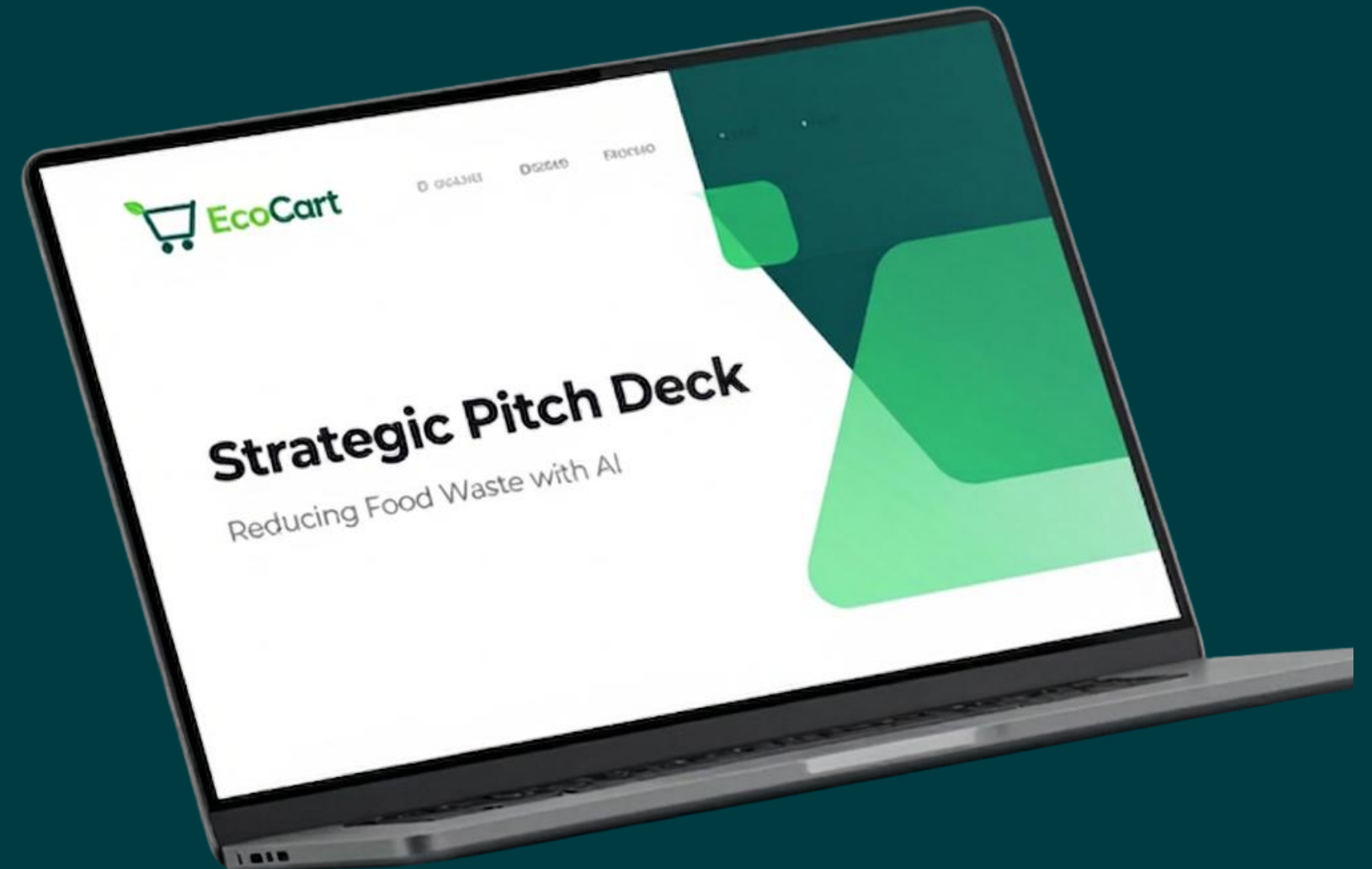
Artifact 3

Artifact 3: Strategic Pitch Deck

(Maximum 10 Slides)

A presentation aimed at the EcoCart Founders summarizing your strategy. It must cover:

- **Executive Summary:** The problem, your solution, and how it serves the personas.
- **Competitor Analysis:** A brief look at how this compares to existing food/recipe apps on the market.
- **Technical Ecosystem Map:** A high-level list or block diagram of the technologies needed.
- **Data & Privacy Strategy:** How will you protect user purchasing data and build trust?
- **Go-To-Market (GTM) / Rollout Plan:** How will you market this new feature to drive adoption?



How you might want
to score yourself (40
Points Total)



Category	Criteria
Product Definition (10 Points)	Are the user stories logical and tied to the personas? Are the chosen KPIs measurable and relevant to the business goal?
UX & Design Thinking (10 Points)	Do the wireframes provide a seamless, intuitive user experience? Does the design actively reduce the friction of logging food?
Technical & Commercial Strategy (10 Points)	Is the proposed tech stack realistic for a startup? Is the Go-To-Market plan creative and likely to drive user adoption?
Risk Mitigation (10 Points)	How robustly does your strategy handle the data privacy concerns surrounding AI and user purchasing habits?



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Engineering Sector Skills Project

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Scenario & Client Issue

Scenario

- You are a **Data Engineer** at **AeroGrid**, a rapidly growing renewal energy provider managing a fleet of offshore wind turbines
- To monitor equipment health, they recently installed IoT (Internet of Things) sensors on all their turbines.
- These sensors emit constant telemetry data (continuous readings of timestamp, temperature, RPM, and vibration levels).

Client Issue

- Their current legacy server is struggling to ingest this massive, non-stop flow of data.
- Worse, several turbines have recently failed because the engineering team couldn't spot the warning signs buried in the massive data logs



Your Role

AeroGrid has 4 Platform Engineers focused entirely on keeping the current servers online; they have no time to write **analysis scripts** or **re-architect the system**.

You have been brought in to:

Process a
24-hr
sample of
the data

Find the
failing
machines

Design a
modern,
scalable
cloud system



Deliverables required

You must submit an **Engineering Solutions Package** containing four artefacts.

You have been provided a file named [*telemetry_data.csv*](#) to complete these tasks.

1. Data
processing
code

3. System
Architecture
Diagram

2.
Containerisation
(Dockerfile)

4.
Engineering
report



Artefact 1: Data Processing

Task

Write a runnable, well-commented script (Python is highly recommended) **that parses the provided telemetry_data.csv file.**

It must calculate the metrics and output a clear list of the "Turbine IDs" that are failing the anomaly rules.

Include brief instructions on how to run your code.



What are Anomaly Rules?

In engineering, an **anomaly** is a data point that **deviates** from what is normal or safe.

A turbine requires **urgent maintenance** if:

- Average Temperature exceeds **85.0°C**.
- Vibration levels spike above **15.0 mm/s**.



Hint:

"Parsing" just means reading and analyzing the data. If you are using Python, look into a library called **pandas**

It is the **industry standard** for handling massive spreadsheets and CSV files efficiently

Artefact 2: Containerisation



Task

To ensure your script runs reliably in the cloud environment you are about to propose, write a **Dockerfile** that packages your Python script and any necessary dependencies into a runnable container.



What is a Dockerfile and Containerisation?

The biggest headache in engineering is "It works on my machine, but not on the server!"

Containerisation solves this.

A Docker container is like a standardised shipping container for software - it packages your code, the operating system, and all the required libraries together so it runs exactly the same way everywhere.

A Dockerfile is simply the text document (the recipe) that contains the commands to assemble that container.

Artefact 3: System Architecture Diagram



Task

Create a **visual diagram** showing a proposed cloud infrastructure to handle this streaming data in real time. Move us away from a single-server setup.

- Show how data flows from the turbine sensors, through a processing layer, and into a database.
- You can draw this by hand and scan it, or use free digital tools like draw.io or Lucidchart.

Tips

Modern systems don't send data straight to a database, because if the database crashes, the data is lost. Instead, research these cloud concepts to include in your diagram:

- **Message Queues / Pub-Sub** (e.g., Apache Kafka, AWS Kinesis): A "waiting room" that safely holds data spikes before processing.
- **Stream Processing** (e.g., AWS Lambda): Small bits of code that run your anomaly rules in real-time as data flows through.
- **Data Storage**: Think about separating "Hot Data" (recent data stored in a fast Time-Series database for dashboards) from "Cold Data" (old data stored cheaply in a Data Lake).

Artefact 4: Report

Task

Engineering Report (300 words max)

A short, professional summary addressed to the AeroGrid CTO. It must include:

- **Findings:** Which specific turbines are failing based on your script?
- **Architecture Justification:** Why did you choose the specific cloud components in your diagram? How does it solve the crashing issue?
- **Cost Optimisation:** Suggest one method to optimise cloud computing or storage costs for this massive, continuous data pipeline.



How you might want to score yourself (40 Points Total)



Category	Criteria
Code Quality & Execution (10 Points):	Does the script run successfully? Is the logic mathematically sound and the code clean/commented?
DevOps / Containerisation (10 Points)	Is the Dockerfile structured correctly (e.g., using an appropriate base image, installing dependencies properly)?
System Design (10 Points)	Does the architecture diagram use appropriate industry-standard concepts for high-volume streaming data?
Analytical Communication (10 Points)	Does the written report accurately interpret the data, justify the architecture, and show an understanding of cost-effective cloud engineering?

Top Tips



Open the sector skills project brief – you can find it in the Resources section of the HelpDesk and in your email



A recording of this session will be shared on Zoom Events



You have the next two days to work on the task



If you have questions – ask your BN peers or research online! You're not expected to know everything



Create your work sample in a document that allows you to share it via a link (WeTransfer, Google Drive or Dropbox)



Approach the task as if it were set by a manager in a real work environment



Focus on doing the task - do not worry about submitting the sector skills project



Do your best – we don't grade your work, which means you CANNOT fail the sector skills project



Instructions on how to submit the sector skills project will be shared on day four